EE2213 Introduction to Artificial Intelligence

Lecture 1

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OUR TEACHING TEAM

Shaojing Fan (Instructor)

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Location:

- E4-05-26 (near ECE general office)

Wang Si (Instructor)

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Location:

- E1A-04-01

Leng Yunze (GA---conducts Python Clinic Sessions, handles assignment Q&A and grading)

Email: yleng@u.nus.edu

OVERVIEW OF COURSE CONTENTS

- **Introduction (Shaojing)**
  - What is AI
  - Applications of AI
  - AI agent
- **Search (Shaojing)**
  - Uninformed search algorithms: breadth-first, depth-first, uniform-cost(Dijkstra's algorithm)
  - Informed search algorithms: greedy best-first, A*
  - Applications
- **Optimisation (Shaojing)**
  - Linear programming
  - Convex problems
  - Applications
- **Machine learning (Wang Si)**
  - Supervised and unsupervised learning: regression, classification, clustering
  - Neural networks and deep learning
  - Applications
- **Knowledge representation (Wang Si)**
  - Knowledge Representation and Reasoning
  - Propositional Logic
  - Applications
- **Ethical considerations (Shaojing)**
  - Bias in AI
  - Privacy concerns
  - Societal impact

COURSE PLAN

mechanistic, rule-based forms of intelligence

Search
Optimisation

lower-Level/foundational intelligence

adaptive, data-driven forms of intelligence

Machine learning

mid-level/adaptive intelligence

abstract, reasoning-based forms of intelligence

Knowledge representation

higher-level/reasoning Intelligence

Prerequisites

- Programming (Python)*
- Basic linear algebra & Probability†

*Two optional python clinic sessions will be provided

†Refer to general familiarity rather than specific knowledge

References: https://stanford-cs221.github.io/spring2023/modules/index.html

Grading:

In-Class Quizzes	20%
Projects	20%
Assignments	20%
Final exam	40%

Class Venue:

- **Monday**: LT4
- **Friday**: LT7

Four In-Class Quizzes

No.	Test Content	Grade %	Week	Date
Quiz 1	Search	5%	Week 3	29 Aug 2025
Quiz 2	Optimization	5%	Week 5	12 Sep 2025
Quiz 3	Machine learning	5%	Week 9	17 Oct 2025
Quiz 4	Knowledge representation	5%	Week 11	31 Oct 2025

- Mode: Digital exam conducted via Examplify
- Open-book exam: You may bring any printed or digital materials, but internet access is not allowed. You will have 2 hours to complete the exam.

Gentle reminder: Please lock the dates in your calendar. ***Only students with a valid MC will be excused****, and a make-up quiz will be arranged.*

Reference of Examplify:
https://nus.atlassian.net/wiki/spaces/DAstudent/pages/22511650/Getting+Started

One Project (Individual):

Machine learning 20%

- This is an open-book project. You will be given four weeks to complete it, and all coding must be done in Python.

- Late submission policy:
  - 0–12 hours late: 80% of your grade
  - 1 day late: 50% of your grade
  - 2 days late: 30% of your grade
  - After 2 days: 0

Four After-Class Assignments

Search (2 assignments)	4% + 6%
Optimization (1 assignments)	5%
Knowledge Representation (1 assignments)	5%

- Open-book; you will have 2–3 weeks to complete each assignment.
- Code will be autograded, with feedback provided on a subset of test cases.
- Late submission policy:
    - 0–12 hours late: 80% of your grade
    - 1 day late: 50% of your grade
    - 2 days late: 30% of your grade
    - After 2 days: 0

THE HONOR CODE

- Do collaborate and discuss together, but write up and code independently.
- Do not look at anyone else's writeup or code.

Python Clinic Sessions

Online Python Clinic Session 1 for EE2213:

- Slot 1: Thursday, 21 Aug, 9:00–10:00 PM
- Slot 2: Monday, 25 Aug, 3:00–4:00 PM

Online Python Clinic Session 2 for EE2213:

- Slot 1: Monday, 29 Sep, 3:00–4:00 PM

**Python clinic sessions are optional but highly recommended. They cover fundamental Python concepts and skills to support you in completing assignments and coding-related exam/quiz questions.*

Final Exam (40%)

Goal: To assess your ability to apply knowledge to new problems, rather than just recall facts.

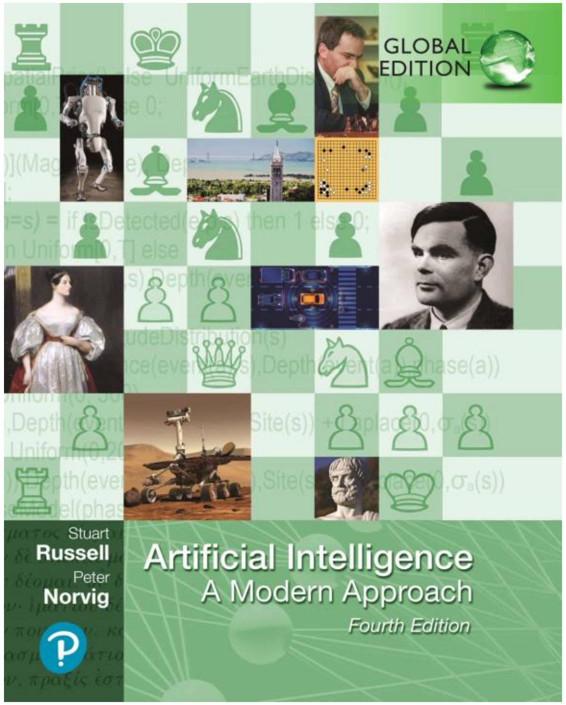
Mode: Digital exam conducted via Examplify

Open-book exam: You may bring any printed or digital materials, but internet access is not allowed.

REFERENCES

Recommended reference book*:
"Artificial Intelligence: A Modern Approach" by Stuart Russell and Peter Norvig

*This is considered the "bible" of AI textbooks



Recommend online resources:
Harvard CS50's Introduction to Artificial Intelligence with Python: https://pll.harvard.edu/course/cs50s-introduction-artificial-intelligence-python

Machine learning specialization by Andrew Ng: https://www.youtube.com/watch?v=vStJoetOxJg&list=PLkDaE6sCZn6FNC6YRfRQc_FbeQrF8BwGI

Any questions about course logistics?

Some Housekeeping

- **Office Hour**: Friday 10.30-11.30am (Week 2-5, Week 12-13)
Feel free to stop by anytime during office hours with questions.

- You're also welcome to ask questions anytime during class.

- If you need additional consultation to catch up during the semester, please don't hesitate to email me. We can arrange a meeting either in person or via Zoom.

- You can reach me through Canvas messaging or email at fanshaojing@nus.edu.sg. If you don't hear back within 48 hours (excluding weekends and public holidays), please kindly send a reminder.

Acknowledgement

The course slides are for teaching use only and include adapted materials from web sources, the book *Artificial Intelligence: A Modern Approach* by Russell and Norvig, and materials by David J. Malan, Brian Yu, Chris Callison-Burch, Rohan Paul, Sriram Sankaranarayanan, Fei Fang, Boyd and Vandenberghe, and others.

Agenda

- A brief history of AI and its current landscape
- What is an AI agent?
- ReAct agents and the rise of agentic AI

What is Artificial Intelligence (AI)?

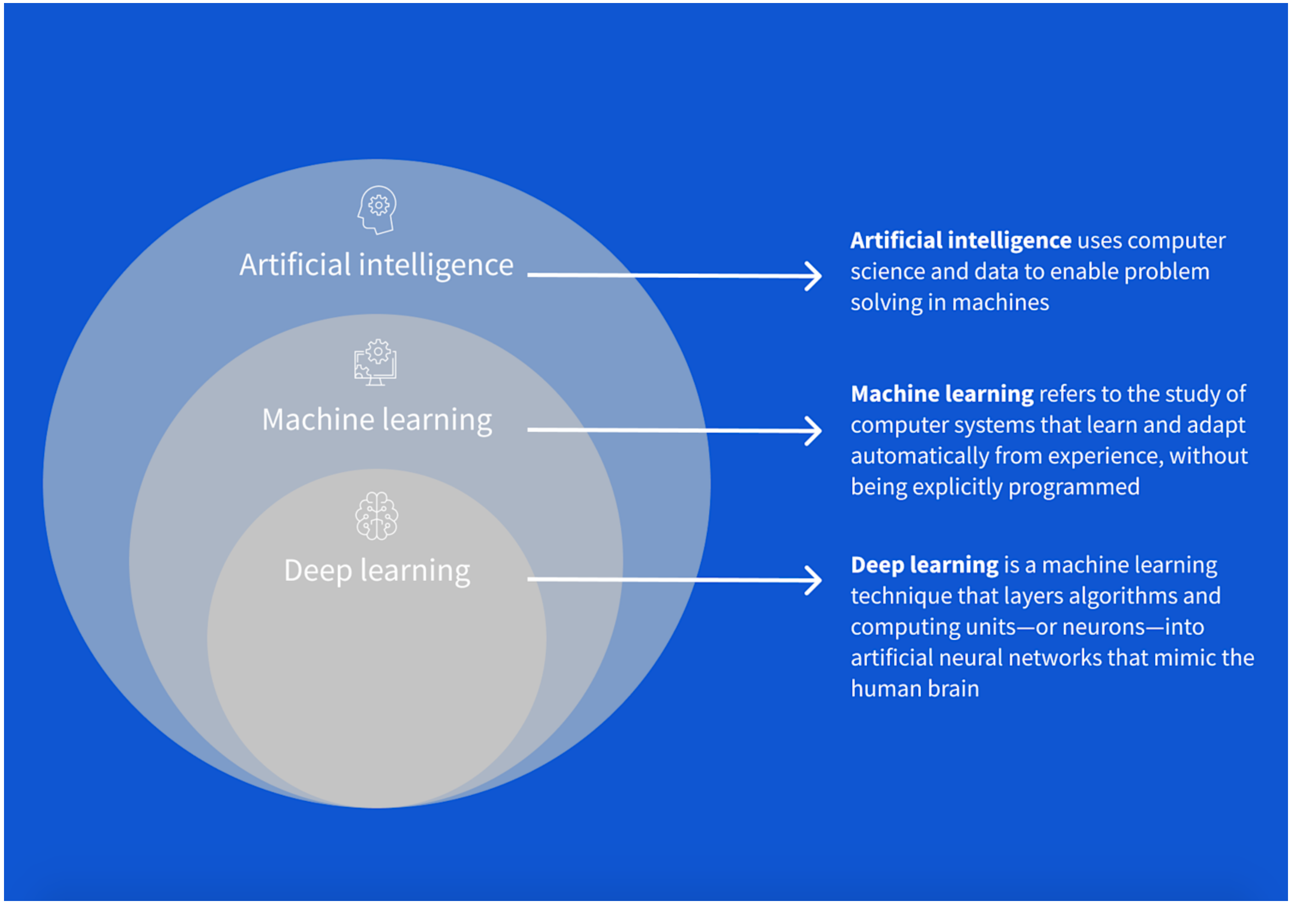
Artificial intelligence is the study of agents that perceive their environment and take actions that maximize their chances of achieving their goals.

— Stuart Russell & Peter Norvig, *Artificial Intelligence: A Modern Approach* (4th ed.)

The science and engineering of making intelligent machines, especially intelligent computer programs.

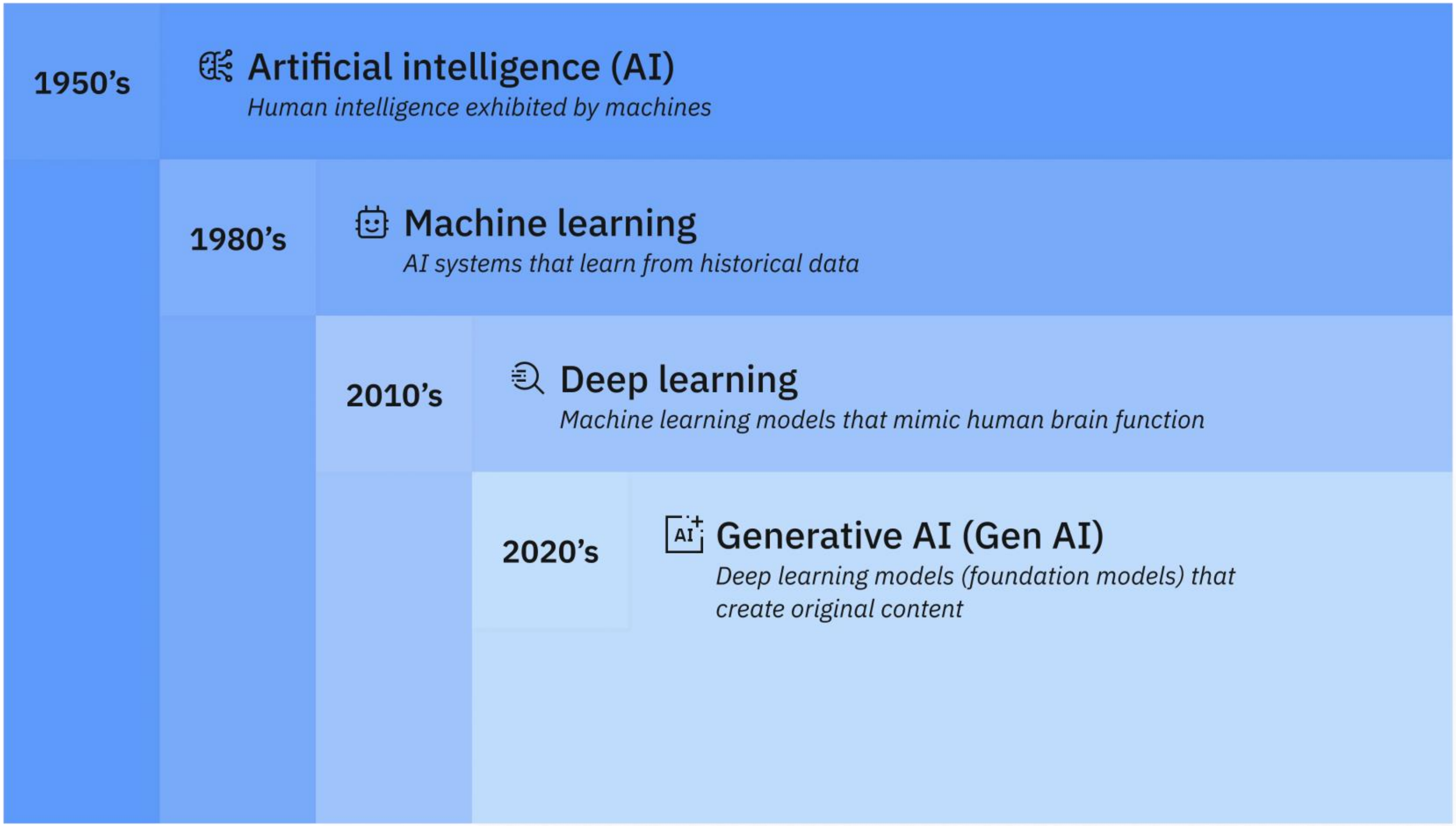
— John McCarthy (1956), the “Father” of AI, who coined the term Artificial Intelligence

AI, Machine Learning, and Deep Learning



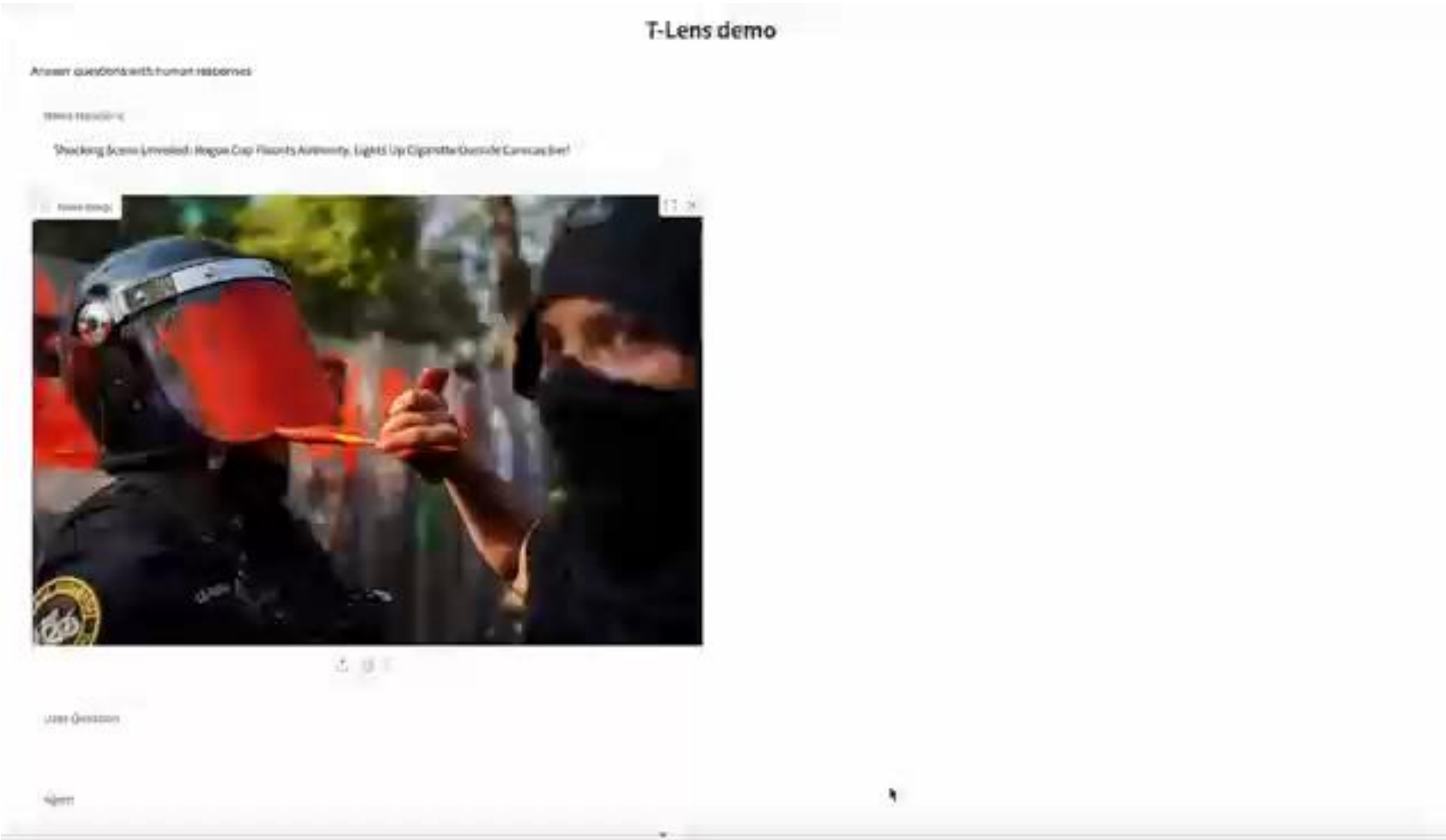
Reference: https://www.coursera.org/articles/ai-vs-deep-learning-vs-machine-learning-beginners-guide

AI, Machine Learning, Deep Learning, and Generative AI



Credit: https://www.ibm.com/think/topics/artificial-intelligence

Why AI?

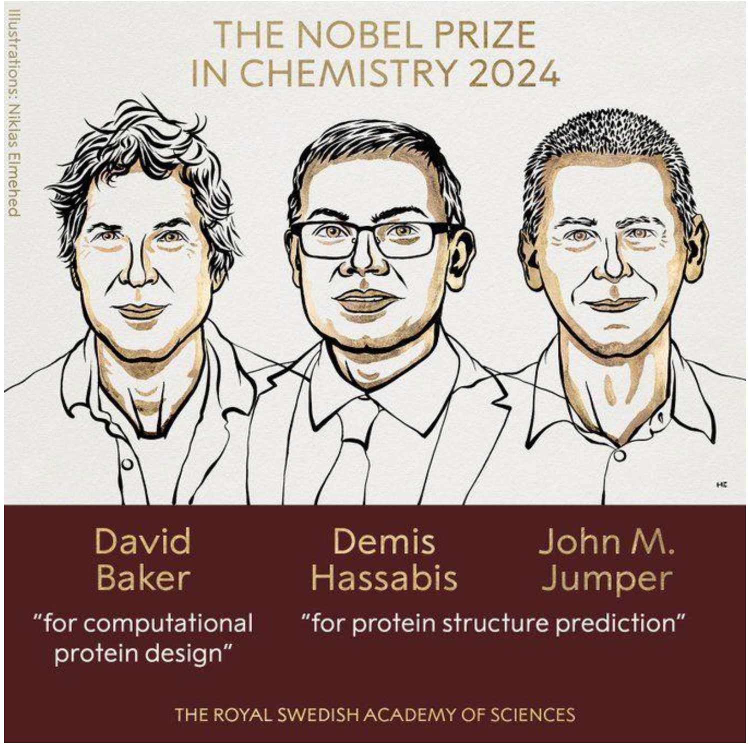


Zhiqi Shen*, Shaojing Fan*, Danni Xu, Terence Sim, and Mohan Kankanhalli. "Real or Fake? Exploring Human Perception of AI-Generated Content", Submitted to AAAI'26

AI and Noble Prize

2024 Nobel prize in Chemistry for *Alphafold*.

A historic milestone: for the first time, an **AI-powered method** has been directly recognized by the Nobel Committee in Chemistry

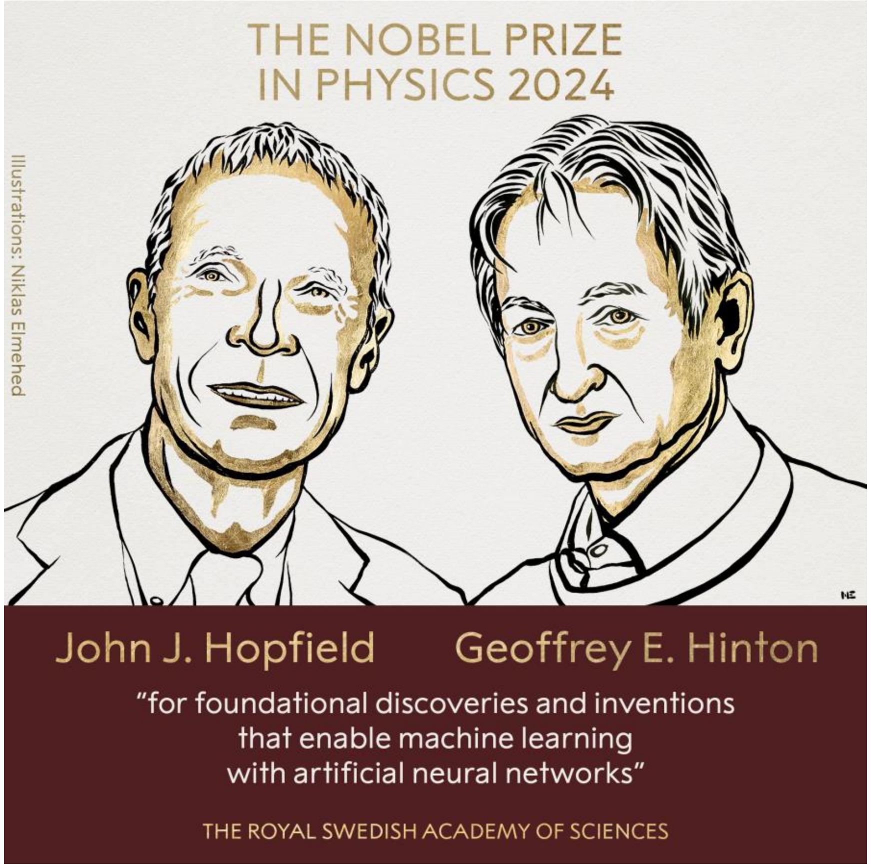


News: https://www.theverge.com/2024/10/9/24265962/google-deepmind-ai-nobel-prize-winners-chemistry-protein?utm_source=chatgpt.com
News: DeepMind's new AlphaGenome AI tackles the 'dark matter' in our DNA, https://www.nature.com/articles/d41586-025-01998-w, June2025
Z.Avsec, et. al, "AlphaGenome: advancing regulatory variant effect prediction with a unified DNA sequence model", Google DeepMind

Why AI?

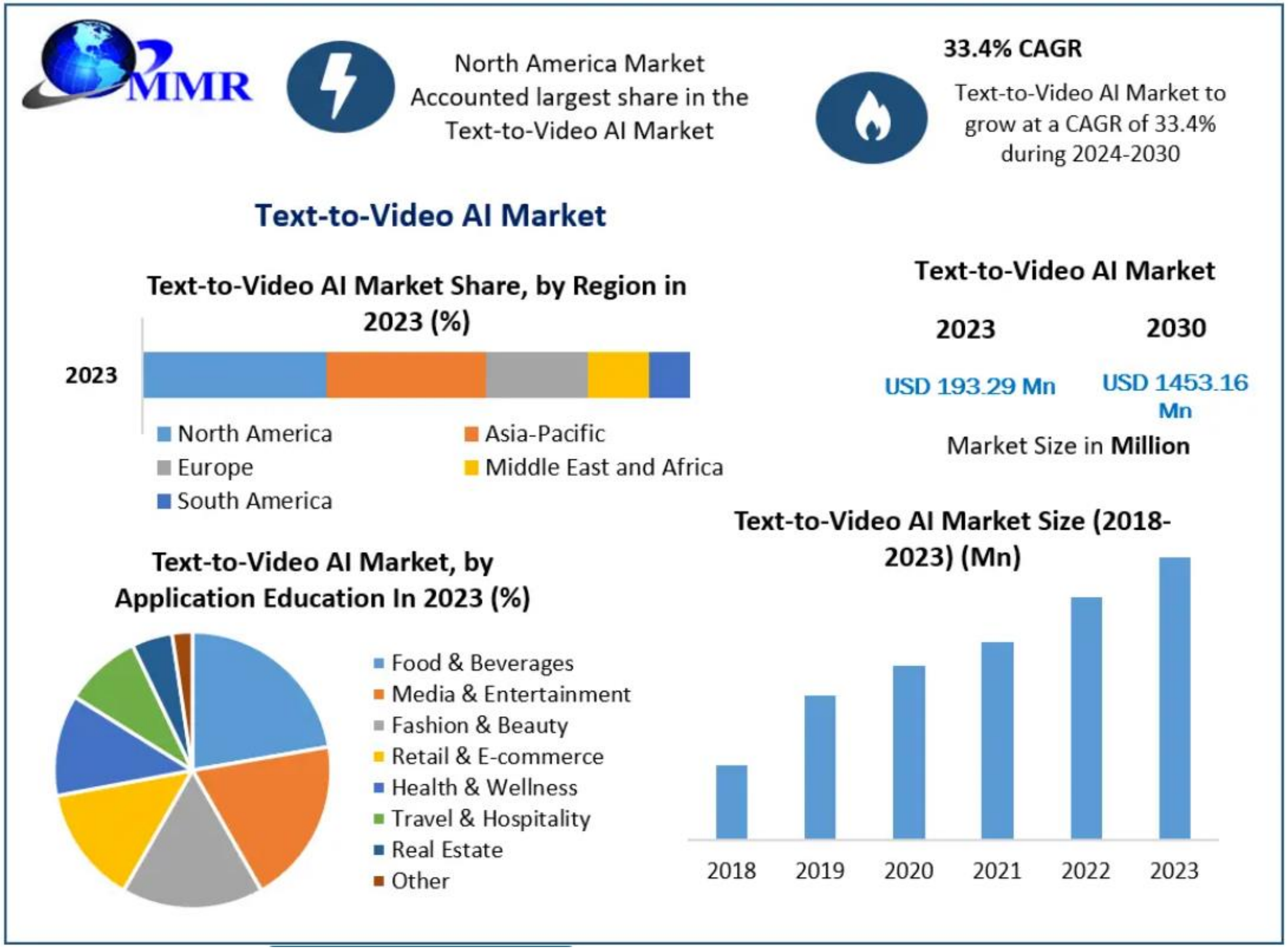
2024 Nobel prize in Physics for artificial neural networks.

A historic milestone: for the first time, the Nobel Prize in Physics has been awarded to researchers primarily working in **computer science and AI**.



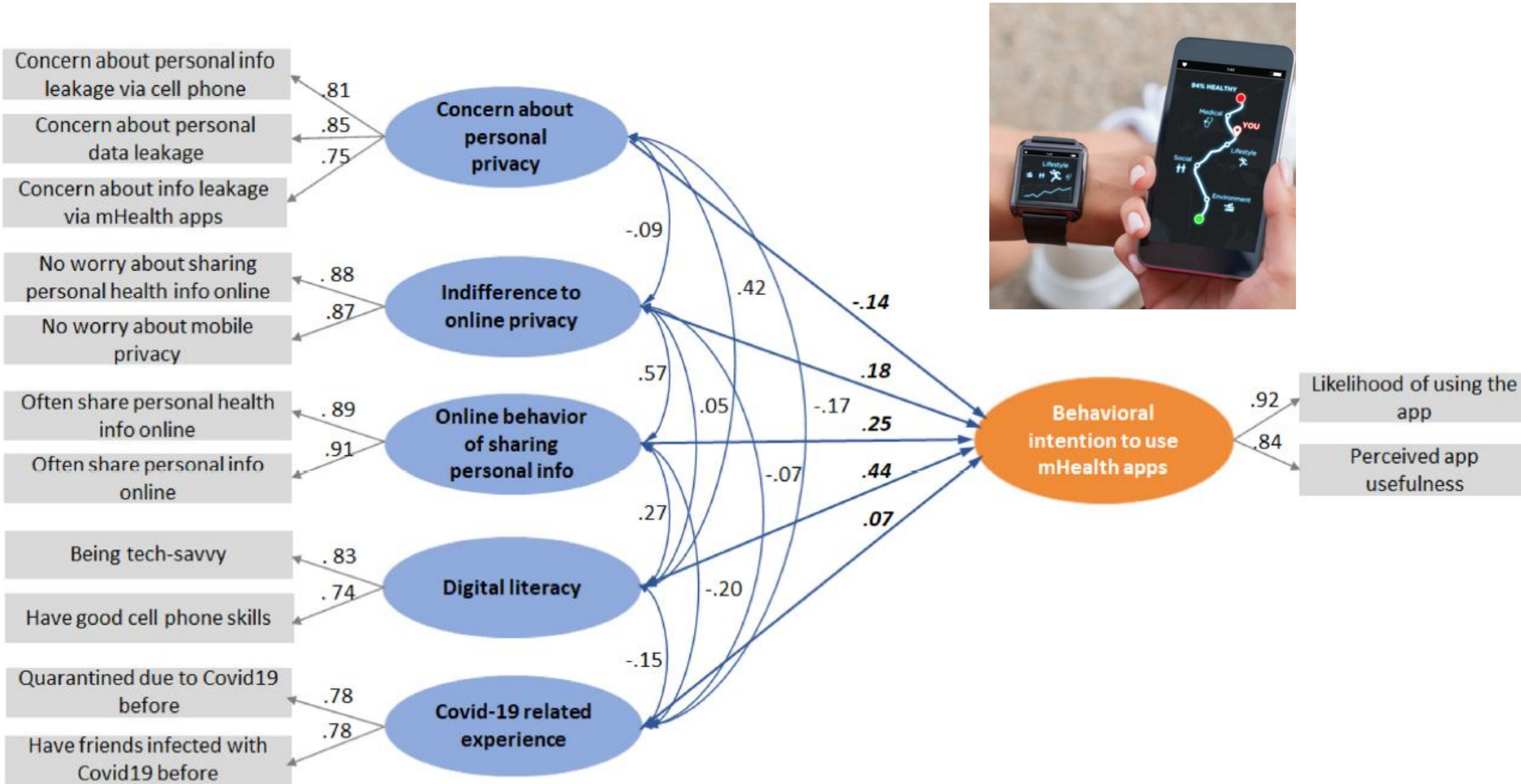
News: https://www.theverge.com/2024/10/9/24265962/google-deepmind-ai-nobel-prize-winners-chemistry-protein?utm_source=chatgpt.com

AI Market



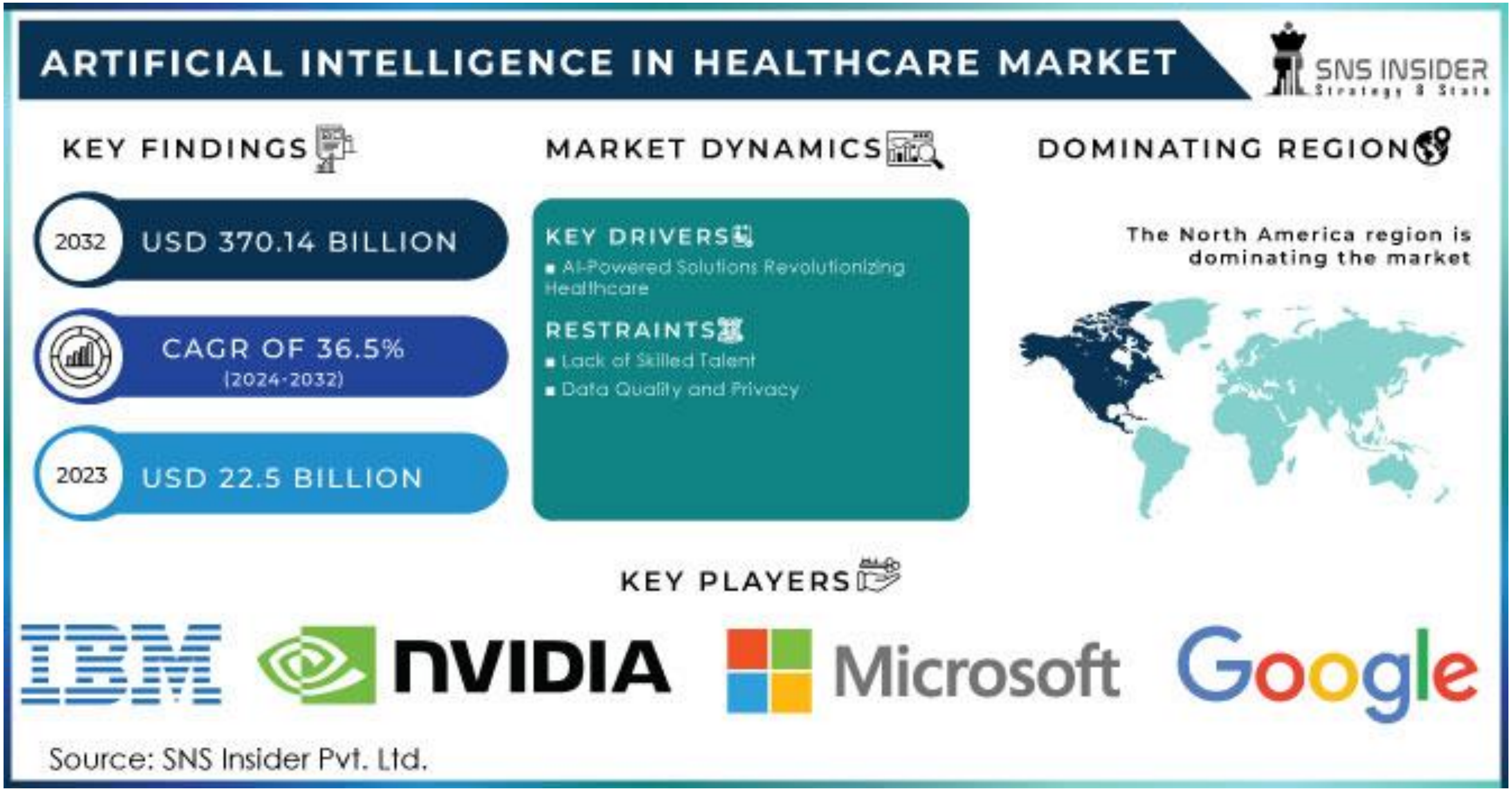
Credit: https://www.maximizemarketresearch.com/market-report/text-to-video-ai-market/190642/

AI for Healthcare



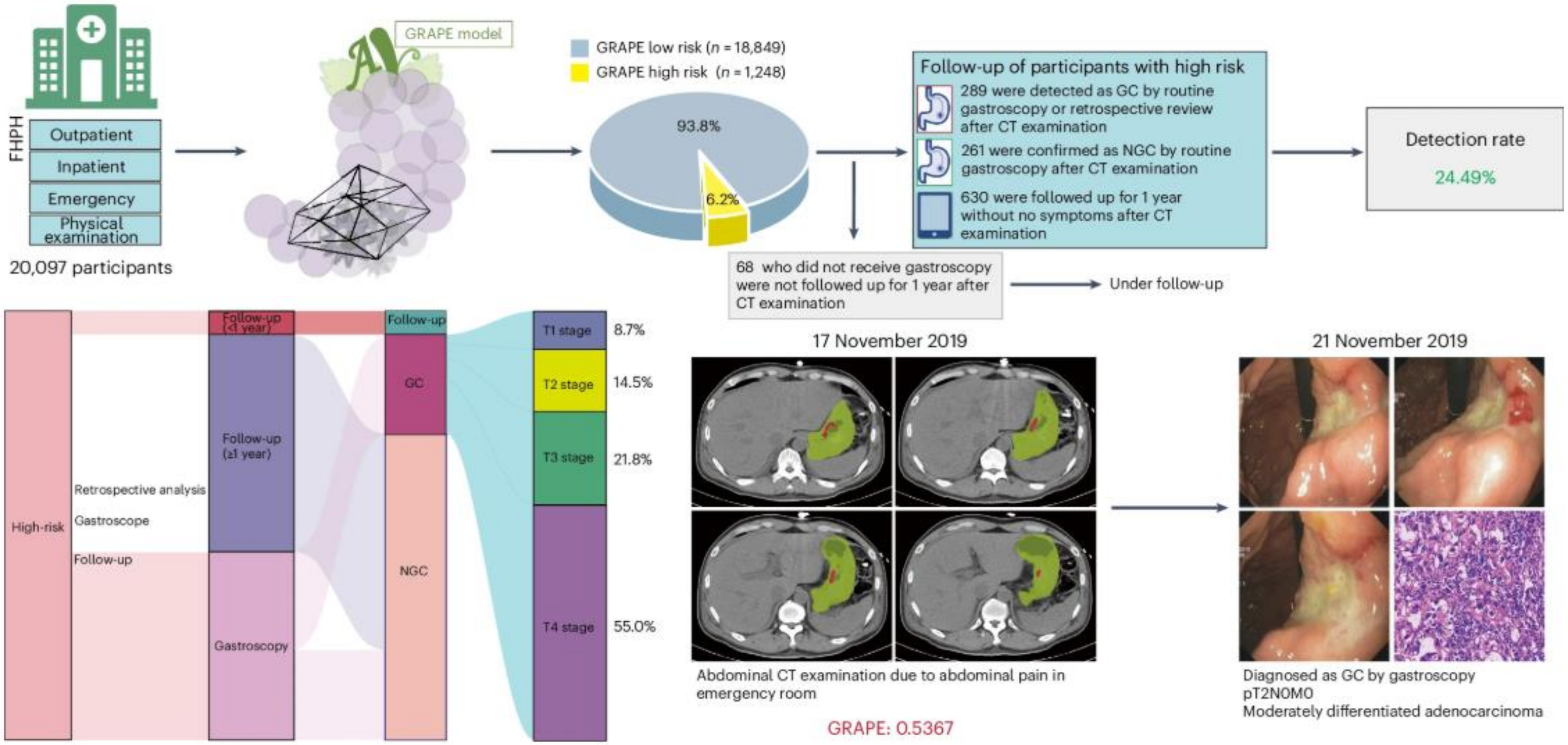
Fan, Shaojing, Ramesh C. Jain, and Mohan S. Kankanhalli. "A comprehensive picture of factors affecting user willingness to use mobile health applications." ACM Transactions on Computing for Healthcare 5.1 (2024): 1-31.

AI for Healthcare



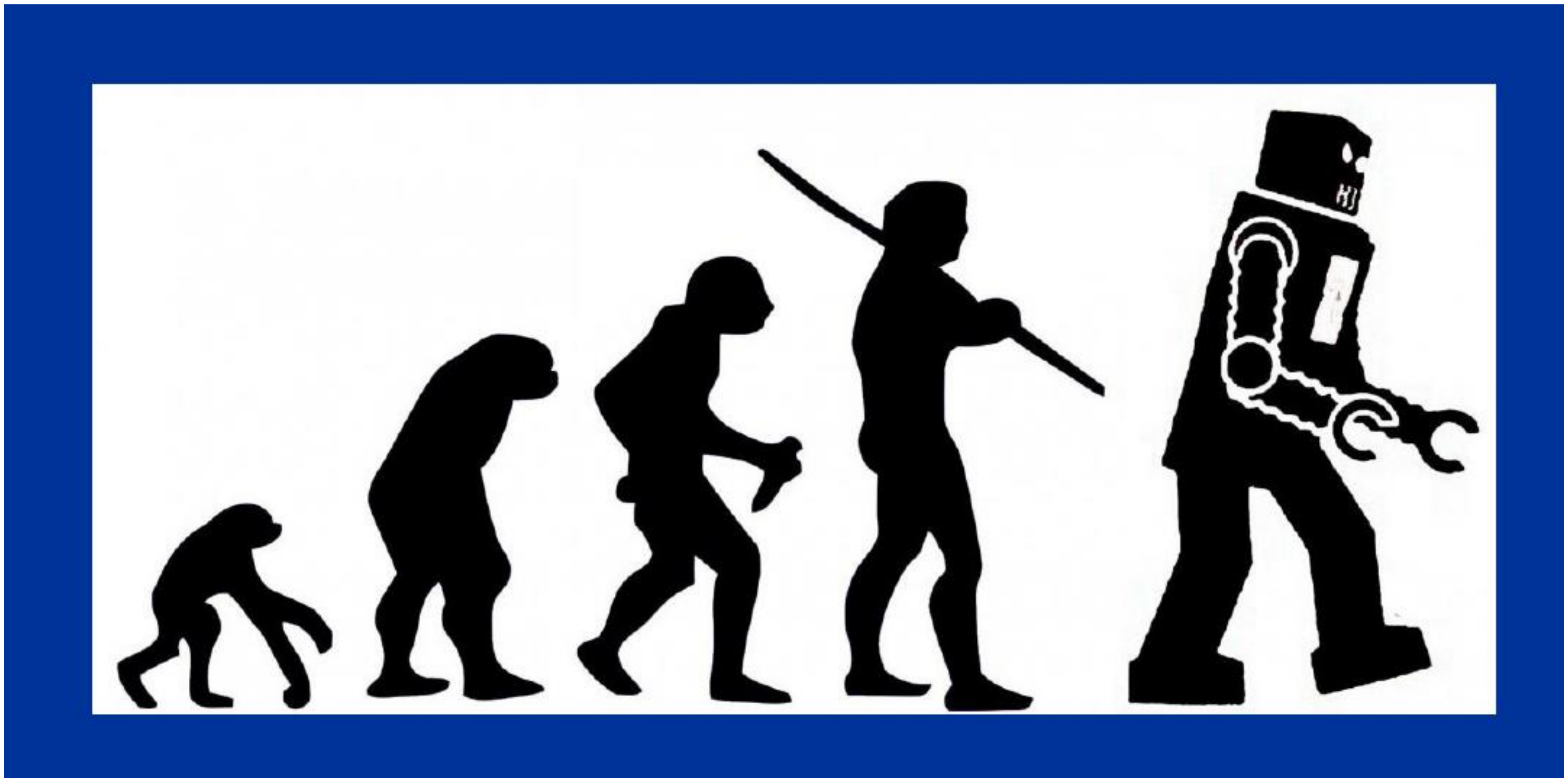
Credit: https://www.snsinsider.com/reports/artificial-intelligence-in-healthcare-market-2278

AI for Healthcare

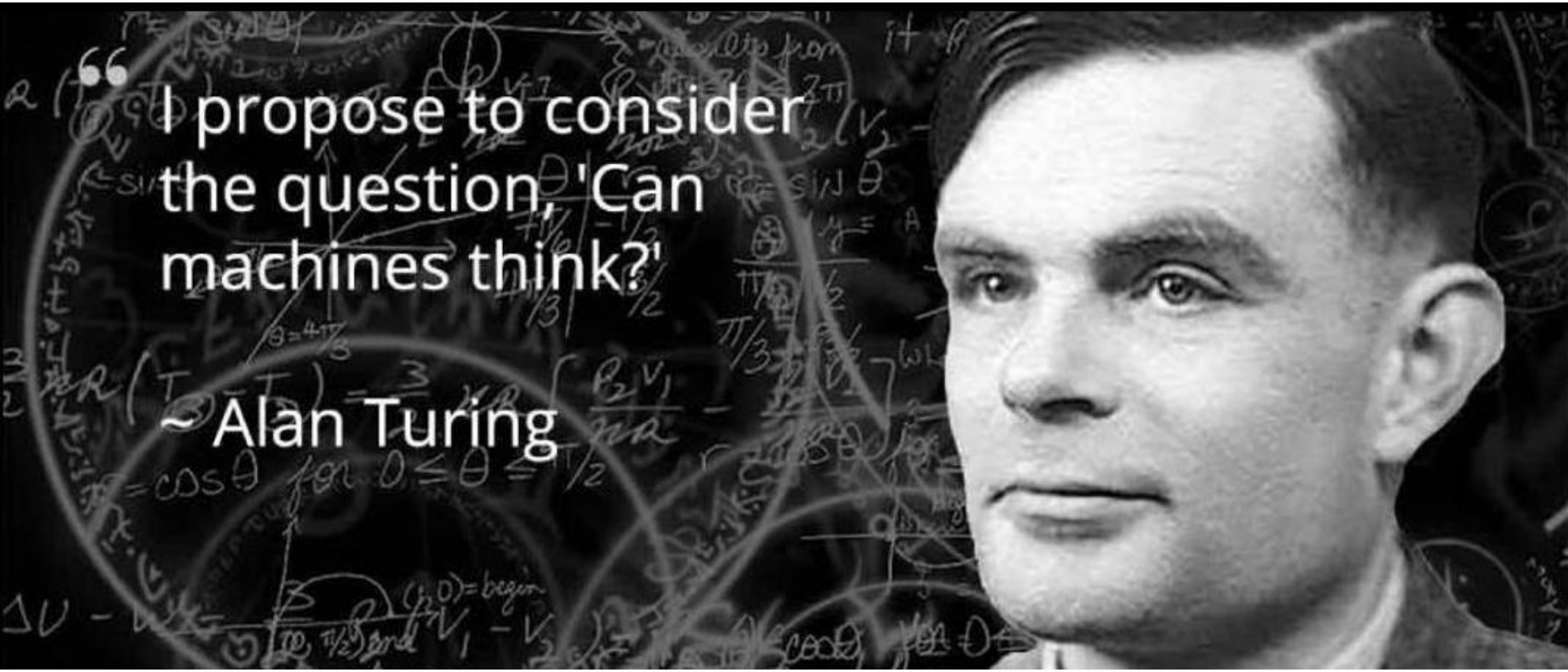


Hu, Can, et al. "AI-based large-scale screening of gastric cancer from noncontrast CT imaging." Nature Medicine (2025): 1-9.

A Brief History of AI



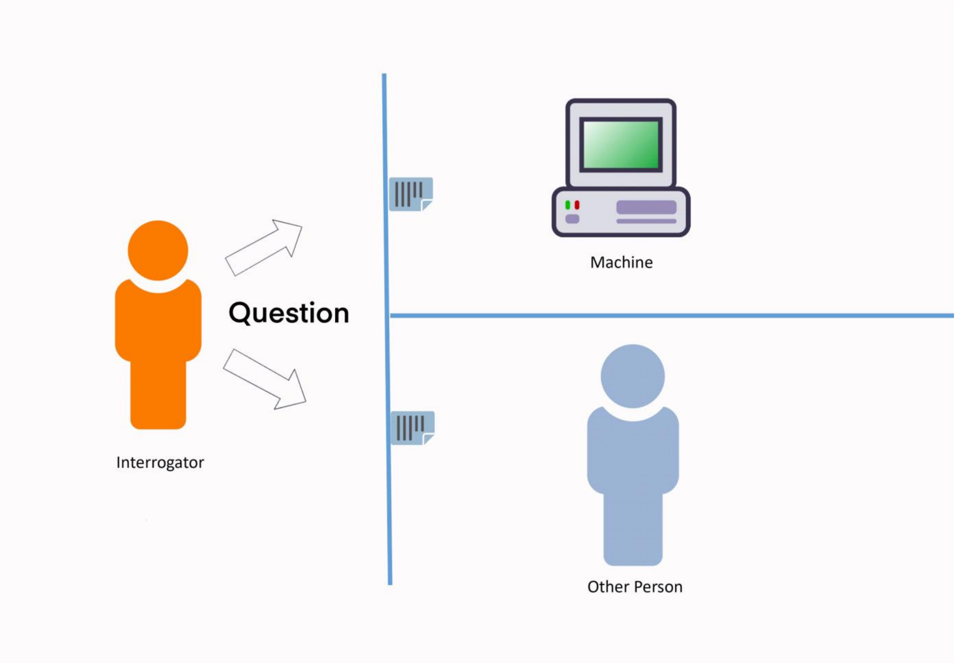
Alan Turing and Turing Test



Alan Turing and Turing Test

“A computer would deserve to be called intelligent if it could deceive a human into believing that it was a human.”

— Alan Turing



Reference: Jones, Cameron R., and Benjamin K. Bergen. "Large language models pass the turing test." *arXiv preprint arXiv:2503.23674* (2025).

Turing Award

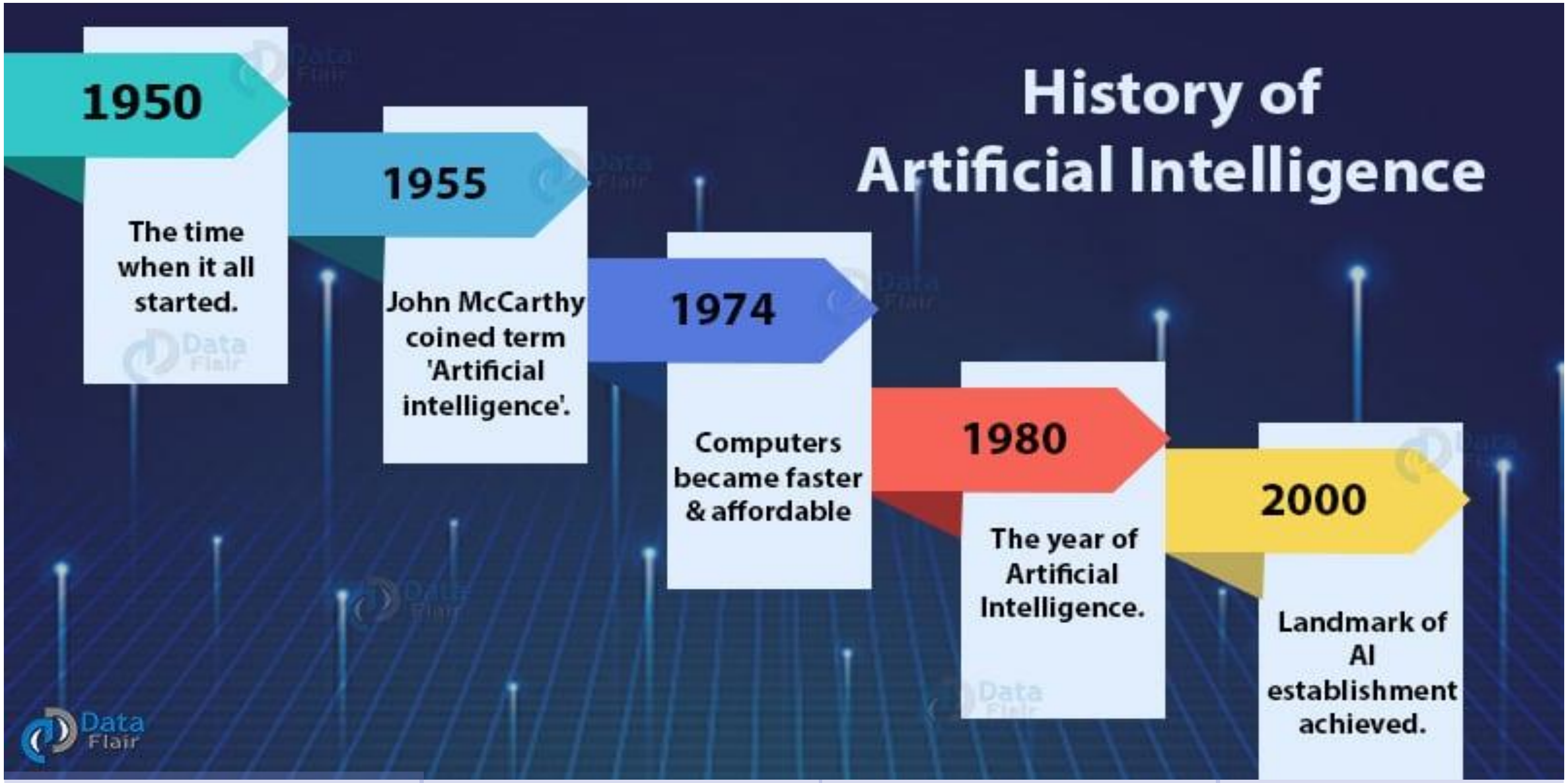
The top award in computer science, including the field of artificial intelligence (AI)

- (2024) Barto, Andrew; Sutton, Richard
- (2023) Wigderson, Avi
- (2022) Metcalfe, Robert Melancton
- (2021) Dongarra, Jack
- (2020) Aho, Alfred Vaino; Ullman, Jeffrey David
- (2019) Catmull, Edwin E. ; Hanrahan, Patrick M.
- (2018) Bengio, Yoshua; Hinton, Geoffrey E; LeCun, Yann



Reference (NUS120 Distinguished Speaker Series | Professor Yoshua Bengio): https://www.youtube.com/watch?v=luJxOyryJ0o

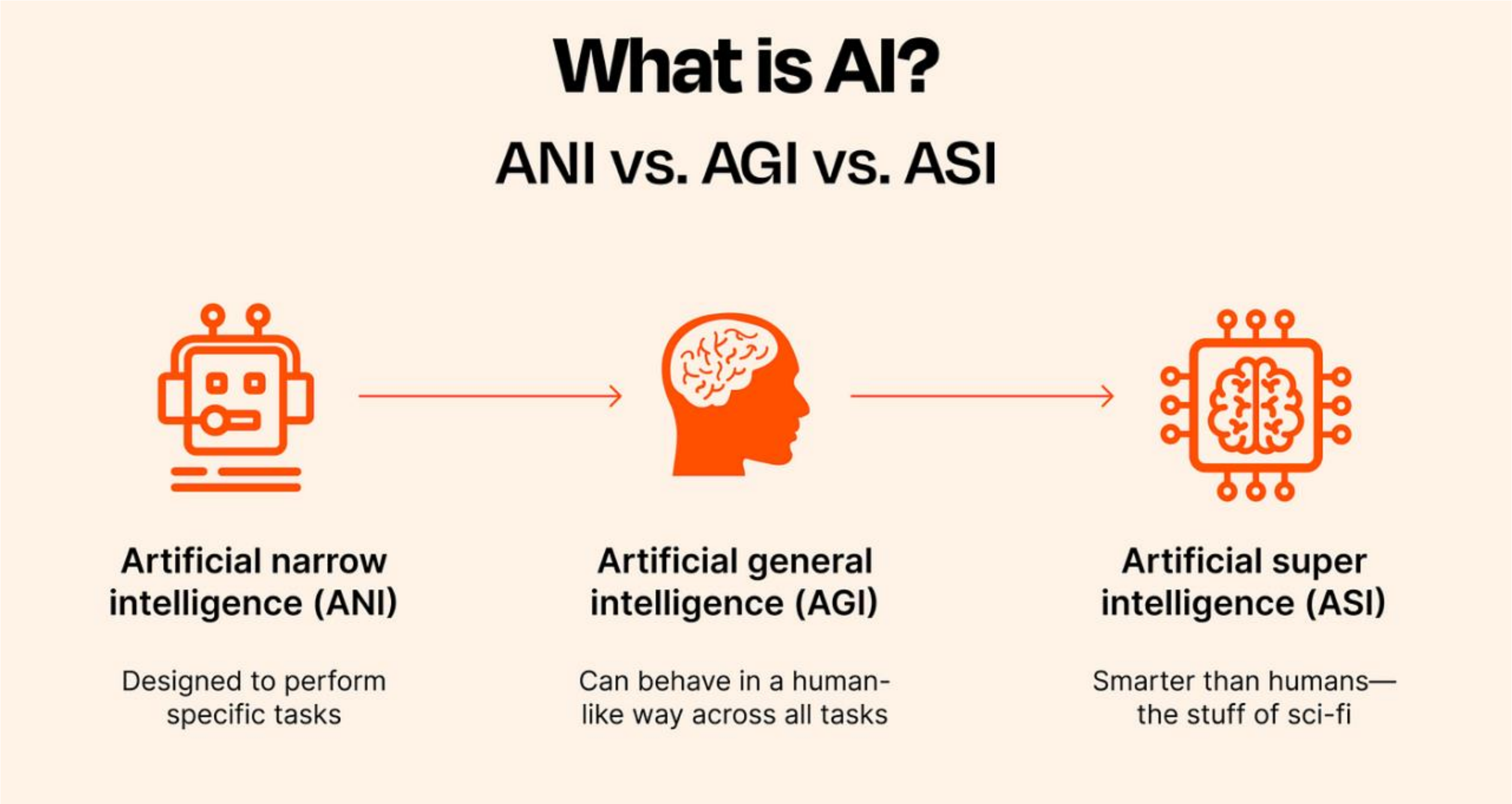
A Brief AI History



Birth of AI (1950s)	Early Development (1960s-70s)	Resurgence (1980s-2000s)	Modern AI (2010s-Present)
Turing Test (1950) Dartmouth Conference (1956)	ELIZA chatbot First AI Winter	Rule-based systems Deep Blue (1997)	Deep learning Generative AI

Reference: https://data-flair.training/blogs/history-of-artificial-intelligence/

AI in Future



Nick, Bostrom. "Superintelligence: Paths, dangers, strategies." 2014,

AI in Future

AI STAGES	ANI	AGI	ASI
Stage	Past & Present	Present-in-progress	Future
Type	Narrow/Single-task	General/Human-level	Superhuman
Scope of Intelligence	Specific Domains only	Multi-domain understanding	All-encompassing, self-improving
Development Status	Already achieved	In active progress	Still Theoretical
Risk Level	Low	Moderate	Not yet realized
Human Comparison	A skilled specialist	A well-rounded thinker	High(speculative)
Examples	Google Maps, Netflix recommendations	Experimental LLMS	Beyond the limits of human capability

https://tutorialsdojo.com/agi-vs-asi-vs-ani-ai-stages-towards-super-intelligence/

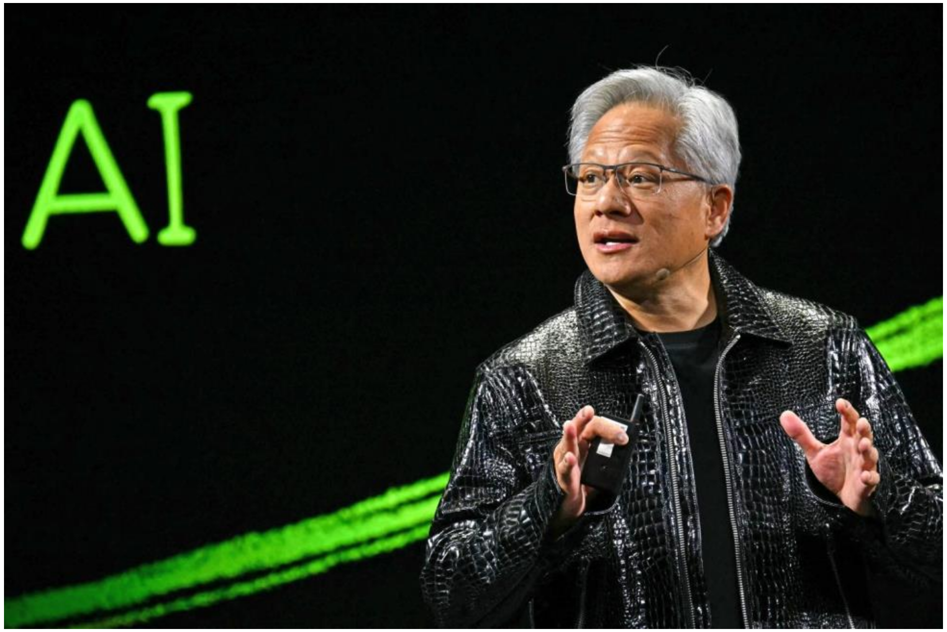
Agenda

- A brief history of AI and its current landscape
- What is an AI agent?
- ReAct agents and the rise of agentic AI

Agentic AI (2025)

"2025 is the year of agentic AI."

— Jensen Huang, CEO of NVIDIA



References: https://www.barrons.com/articles/nvidia-stock-ceo-ai-agents-8c20ddfb
Andrew Ng, "The Rise Of AI Agents And Agentic Reasoning", BUILD 2024 Keynote.

Example of AI Agent

LandingAI

manus
The general AI agent

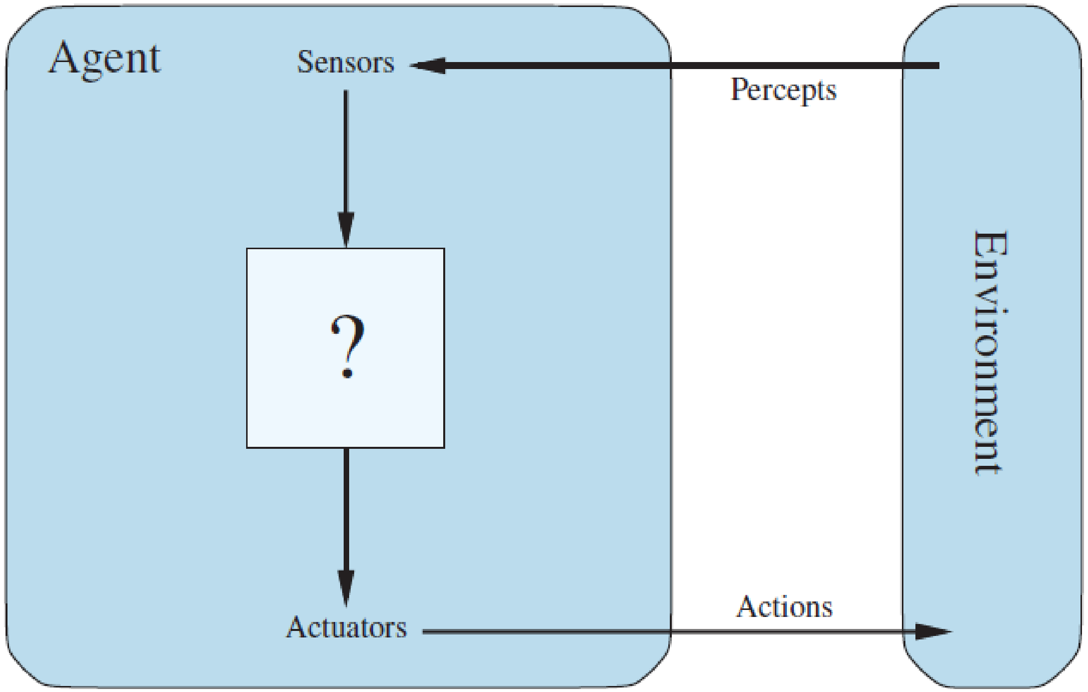
OpenAI Operator
1st AI Agent by OpenAI

Definition of Agent

An **agent** is anything that **perceives** its environment through **sensors** and can **act** on its environment through **actuators**

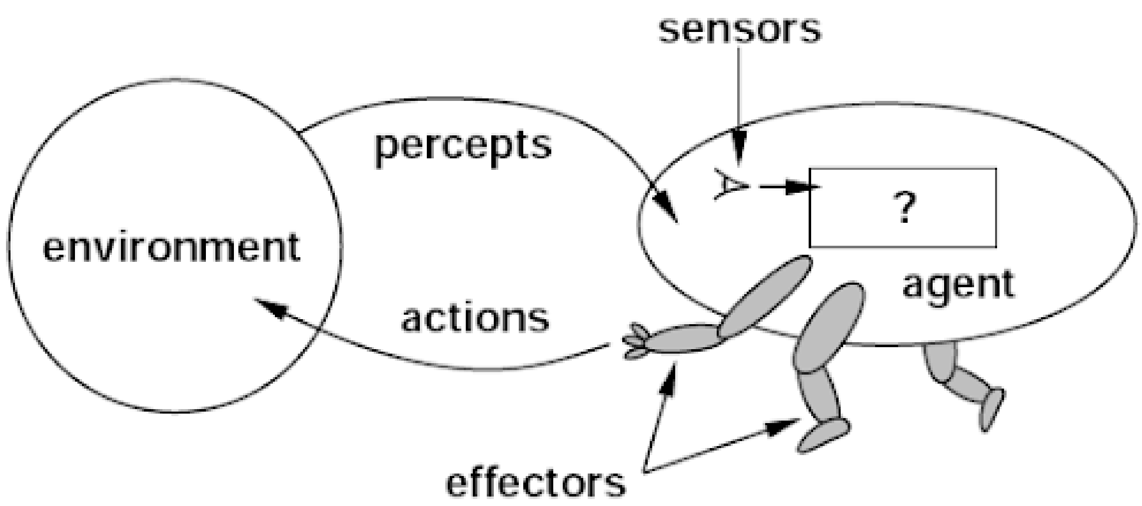
A **percept** is the agent's perceptual inputs at any given instance

An **actuator** is the part of an agent that allows it to take action in the environment based on its decisions.



Reference: Stuart Russell & Peter Norvig, *Artificial Intelligence: A Modern Approach* (4th ed.) P54 – P57.

Agents and Environments



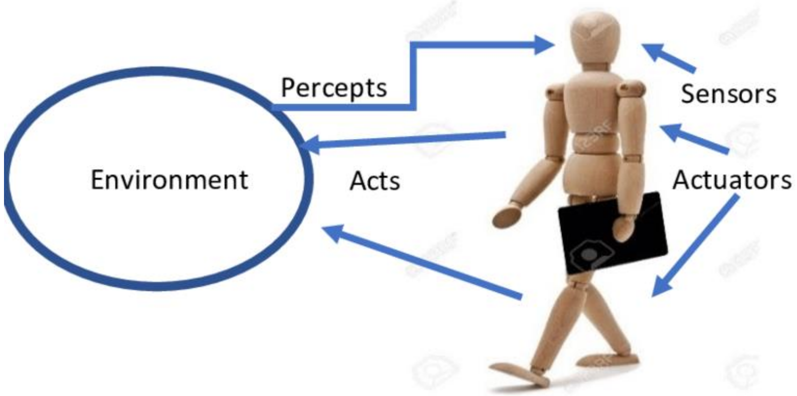
An agent is specified by an *agent function* $f: P \rightarrow a$ that maps a sequence of percept vectors P to an action a from a set A:

$$P = [p_0, p_1, \dots, p_t]$$
$$A = [a_0, a_1, \dots, a_t]$$

Examples of Agents

An ***agent*** is anything that can be viewed as

- ***perceiving*** its ***environment*** through ***sensors*** and
- ***acting*** upon that environment through ***actuators***



Human agent:

- Sensors: eyes, ears, …
- Actuators: hands, legs, mouth, …

Robotic agent:

- Sensors: cameras and infrared range finders
- Actuators: various motors

Agents include humans, robots, softbots, thermostats…

Performance Measure

How do we know if an agent is acting rationally?

- Informally, we expect that it will do the right thing in all circumstances

How do we know if it's doing the right thing?

We define a ***performance measure***:

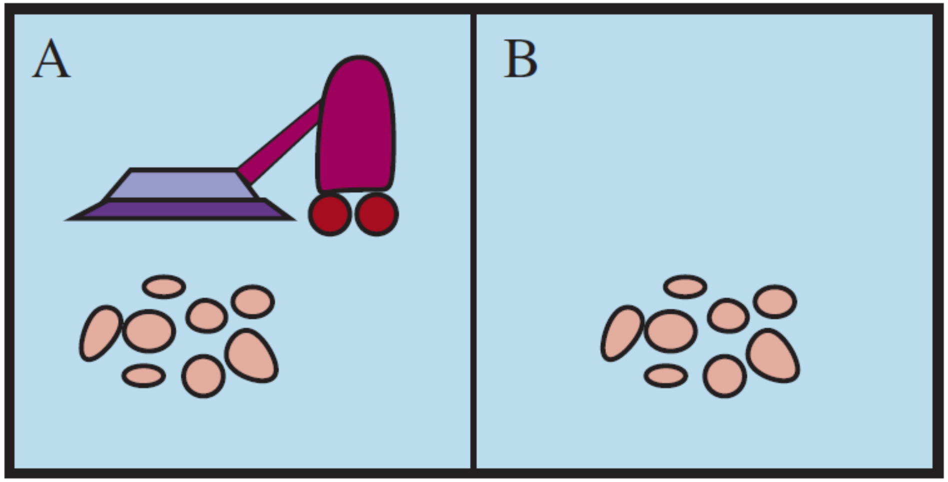
- An objective criterion for success of an agent's behavior
- Given the evidence provided by the percept sequence

E.g., performance measure of a vacuum-cleaner agent could be amount of dirt cleaned up, amount of time taken, amount of electricity consumed, amount of noise generated, etc.

Example of Performance Measure

Performance measure of a vacuum-cleaner agent could be amount of dirt cleaned up, amount of time taken, amount of electricity consumed, amount of noise generated, etc.

- +1 point for each clean square at time T
- -1 point for each move
- -1000 for more than *k* dirty squares

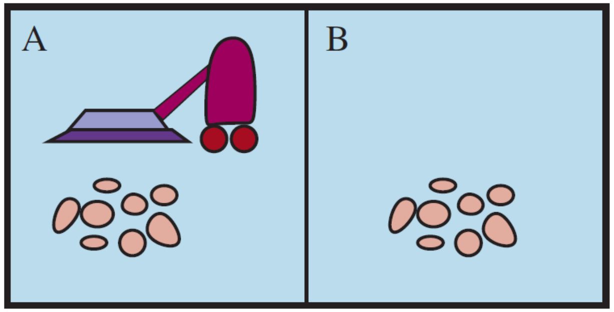


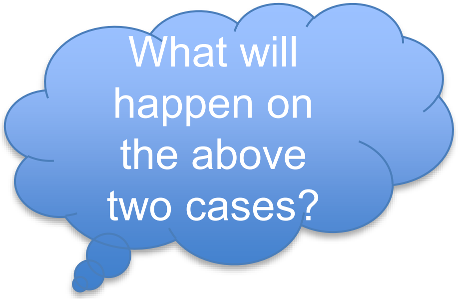
Rule of Thumb for Performance Measure

It is better to design performance measures according to **the goals you want to achieve** in the environment, rather than prescribing **how you think the agent should behave.**

For example, consider these two ways to reward a cleaning robot:

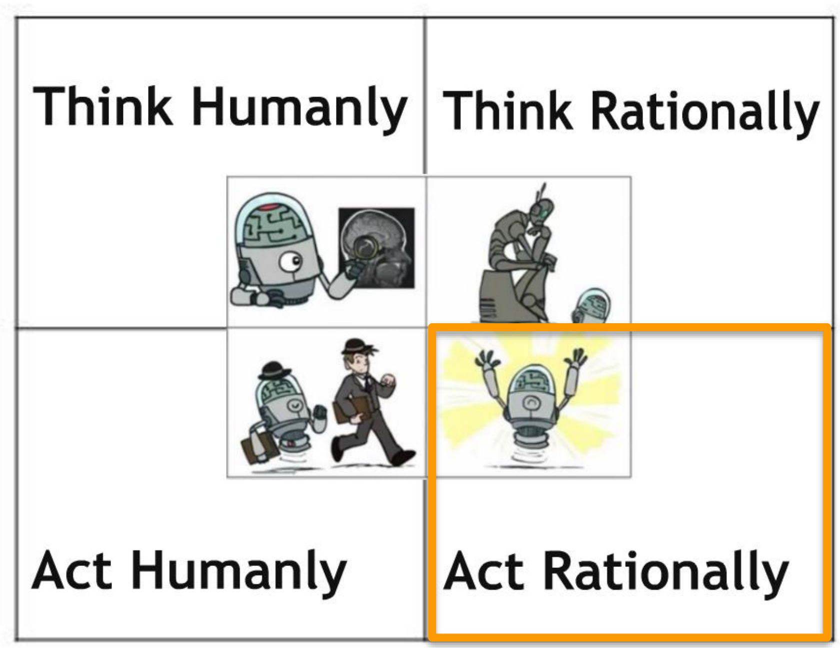
- +1 point for each time the robot cleans a square
- +1 point for each clean square that is clean at a specific time T





Introduction of Rational Agent

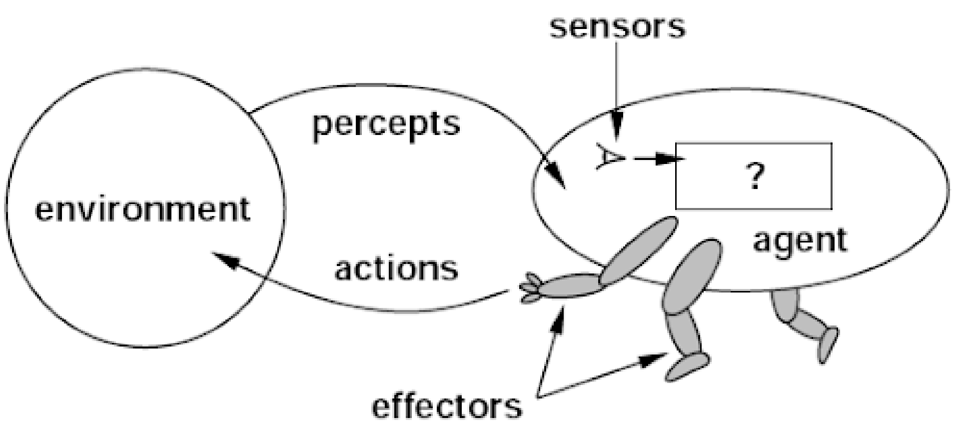
An agent should strive to "do the right thing", based on what it can perceive and the actions it can perform. The right action is the one that will cause the agent to be most successful.



Definition of Rational Agent

Rational Agent:

- For each possible percept sequence *P*
- a rational agent should select an action *a*
- to *maximize* its *performance measure*



Reference: Stuart Russell & Peter Norvig, *Artificial Intelligence: A Modern Approach* (4th ed.) P58.

Rational Agents

- Are rational agents omniscient?

No – they are limited by the available percepts

- Are rational agents clairvoyant?

No – they may lack knowledge of the environment dynamics

- Do rational agents explore and learn?

Yes – in unknown environments these are essential

- Do rational agents make mistakes?

No – but their actions may be unsuccessful / suboptimal

- Are rational agents autonomous (i.e., transcend initial program)?

Yes – as they learn, their behavior depends more on their own experience

Reference: Stuart Russell & Peter Norvig, *Artificial Intelligence: A Modern Approach* (4th ed.) P58 – P60.

Rational Agents

Rational Agent (initial definition):

- For each possible percept sequence *P*
- a rational agent should select an action *a*
- to maximize its performance measure

It doesn't have to know what the actual outcome will be.

Rational Agent (revised definition):

- For each possible percept sequence *P*
- a rational agent should select an action *a*
- to maximize the **expected value** of its performance measure

Reference: Stuart Russell & Peter Norvig, *Artificial Intelligence: A Modern Approach* (4th ed.) P58 – P60.

Task Environment

To design a rational agent, we need to specify a **task environment** --- the setting/context in which the agent will operate and make decisions

- the problem specification the agent is meant to solve

PEAS: to specify a task environment

- ***P***erformance measure
- ***E***nvironment
- ***A***ctuators
- ***S***ensors

Reference: Stuart Russell & Peter Norvig, *Artificial Intelligence: A Modern Approach* (4th ed.) P60 – P61.

PEAS: Example -- Specifying an Autonomous Vehicle

Performance measure
- ?

Environment
- ?

Actuators
- ?

Sensors
- ?

Reference: Stuart Russell & Peter Norvig, *Artificial Intelligence: A Modern Approach* (4th ed.) P61.

PEAS: Example

Agent: automated vehicle

Performance measure

- safe, fast, legal, comfortable, maximize profits

Environment

- roads, other traffic, pedestrians, customers

Actuators

- steering, accelerator, brake, signal, horn

Sensors

- cameras, LiDAR, speedometer, GPS

Reference: Stuart Russell & Peter Norvig, *Artificial Intelligence: A Modern Approach* (4th ed.) P61.

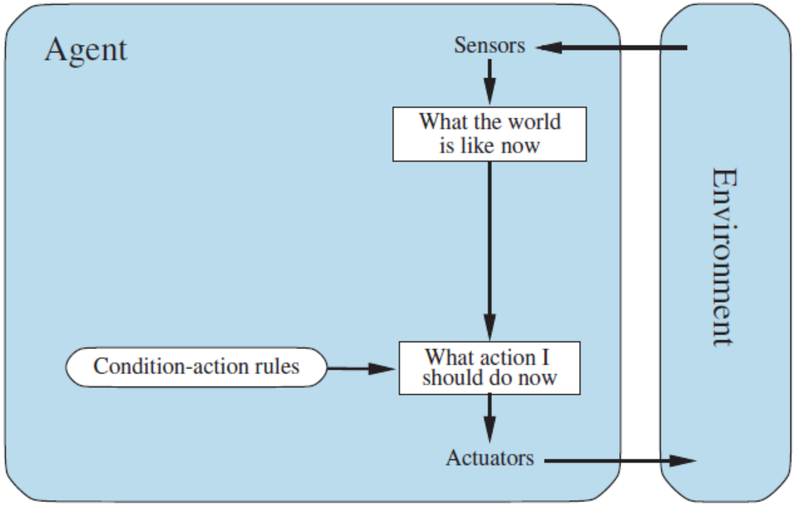
Simple Reflex Agents

- **A Reflex Agent**
  - Selects action based on the current percept.
  - Directly map states to actions.

- Operate using *condition-action* rules.
  - **If** (condition) **then** (action)

- Example:
  - A vacuum cleaner robot that reacts to dirt.
  - **If** *(current-location-is-dirty) then (suck-dirt)*

- Problem: no notion of goals
  - It doesn't plan where to clean next or remember cleaned areas.
  - may repeatedly visit clean spots and miss others.



Reference: Stuart Russell & Peter Norvig, *Artificial Intelligence: A Modern Approach* (4th ed.) P67 – P69.

From Reflex to Problem Solving Agents

- **Reflex Agents**
  - Directly map states to actions.
  - No notion of goals. Do not consider action consequences.

$$x \longrightarrow \boxed{f} \longrightarrow \text{single action } y$$

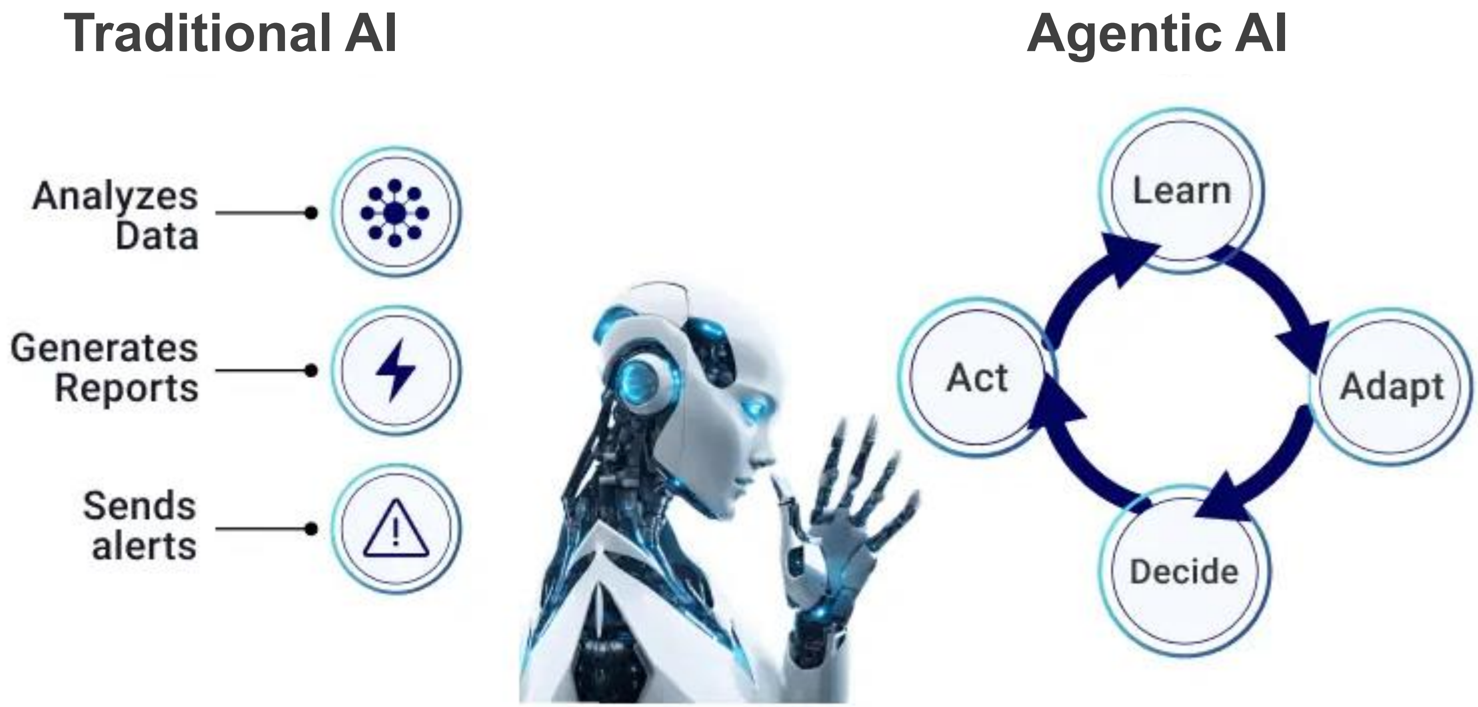
- **Problem Solving Agents**
  - Adopt a goal
  - Consider future actions and the desirability of their outcomes
  - Solution: a sequence of actions that the agent can execute leading to the goal.
  - Today's focus.

$$x \longrightarrow \boxed{f} \longrightarrow \textbf{action sequence } (a_1, a_2, a_3, a_4, \ldots)$$

Agenda

- A brief history of AI and its current landscape
- What is an AI agent?
- ReAct agents and the rise of agentic AI

Traditional AI Systems vs. Agentic AI



Credit: https://www.chooch.com/blog/agentic-ai-for-business-automation-smarter-systems-better-results/

Traditional AI Systems vs. Agentic AI

Features	Traditional AI	Agentic AI
Decision-Making	Follows rules, no autonomy	Independent goal-setting & action
Adaptability	Limited, needs reprogramming	Learns & adjusts continuously
Reactivity	Waits for input	Predicts and acts proactively
Execution	Requires step-by-step guidance	Self-directed execution

Credit: https://www.civo.com/blog/what-is-agentic-ai

Traditional AI Systems vs. Agentic AI

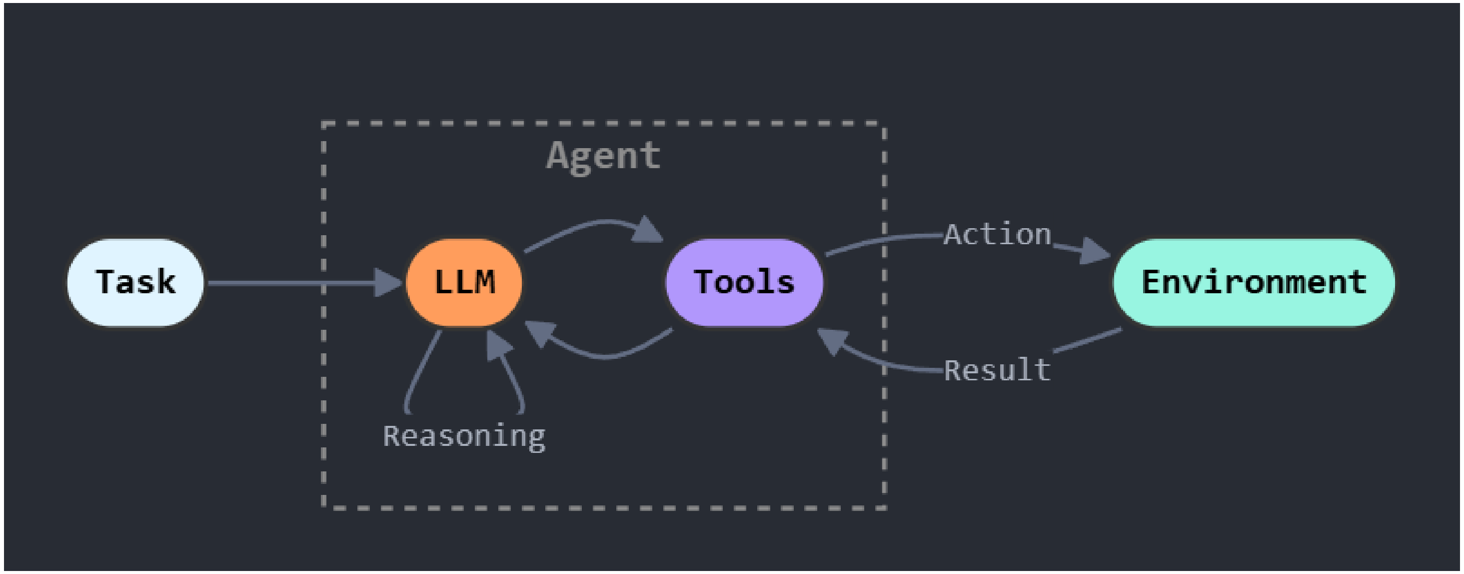
In the era of Large Language Models (LLM):

	Traditional AI systems	Agentic AI
Control logic	Programmatic	LLM
Thinking	Think fast	Think slow

Reference:
https://www.youtube.com/watch?v=F8NKVhkZZWI&list=PL1SWNKG9Fr7kEWZ91kFWw2R3CmXqH7jx1&index=15

ReAct AI Agent

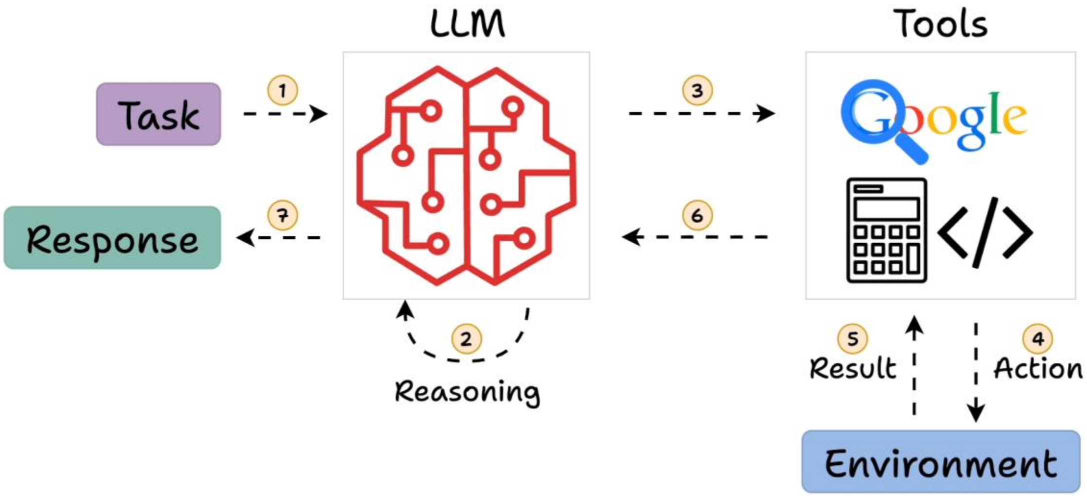
A **ReAct agent** (short for *Reasoning and Acting*) is a framework that enables LLMs to alternately generate explicit reasoning steps (thoughts) and perform actions (like API or tool calls) in a loop—synergizing internal reasoning with external execution to solve complex, multi-step tasks



Reference:
https://medium.com/%40gauritr01/part-1-react-ai-agents-a-guide-to-smarter-ai-through-reasoning-and-action-d5841db39530
https://www.youtube.com/watch?v=F8NKVhkZZWI&list=PL1SWNKG9Fr7kEWZ91kFWw2R3CmXqH7jx1&index=15
https://medium.com/@DanGiannone/demystifying-ai-agents-react-style-agents-vs-agentic-workflows-cedca7e26471

ReAct AI Agent

I am in London, do I need an umbrella today?



- Understand the query (language understanding)
- Select the tools it would need to generate a response (the weather API).
- Gather the weather report.
- Understand if it’s going to rain and produce an answer accordingly.

Reference: https://blog.dailydoseofds.com/p/intro-to-react-reasoning-and-action
https://www.youtube.com/watch?v=F8NKVhkZZWI&list=PL1SWNKG9Fr7kEWZ91kFWw2R3CmXqH7jx1&index=15

References

Definition of AI:
https://www.ibm.com/think/topics/artificial-intelligence
https://www.coursera.org/articles/ai-vs-deep-learning-vs-machine-learning-beginners-guide

Turing Test:
https://www.youtube.com/watch?v=VMPJp7CDuyI
https://www.youtube.com/watch?v=7IG_g3vgDVE

AI agent:
https://www.youtube.com/watch?v=F8NKVhkZZWI&list=PL1SWNKG9Fr7kEWZ91kFWw2R3CmXqH7jx1&index=15

https://www.youtube.com/watch?v=qU3fmidNbJE&list=PL1SWNKG9Fr7kEWZ91kFWw2R3CmXqH7jx1&index=15

Further Reading

Turing's Computing Machinery and Intelligence:
https://www.csee.umbc.edu/courses/471/papers/turing.pdf

History and Philosophy of Neural Networks:
https://research.gold.ac.uk/10846/1/Bishop-2014.pdf

Agentic AI:
https://www.youtube.com/watch?v=KrRD7r7y7NY&list=PL1SWNKG9Fr7kEWZ91kFWw2R3CmXqH7jx1&index=13

Ask about AI (Full version): https://www.youtube.com/watch?v=pe-4W1-mdIQ&list=PLoROMvodv4rPEjA3yzoqkq3J321MfH7FZ&index=6

AI Trends for 2025: https://www.youtube.com/watch?v=5zuF4Ys1eAw

Wikipedia article: https://en.wikipedia.org/wiki/History of articial intelligence

Encyclopedia of Philosophy article: https://plato.stanford.edu/entries/articial-intelligence


Thank you